

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 50

Code of Conduct

I _____ (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

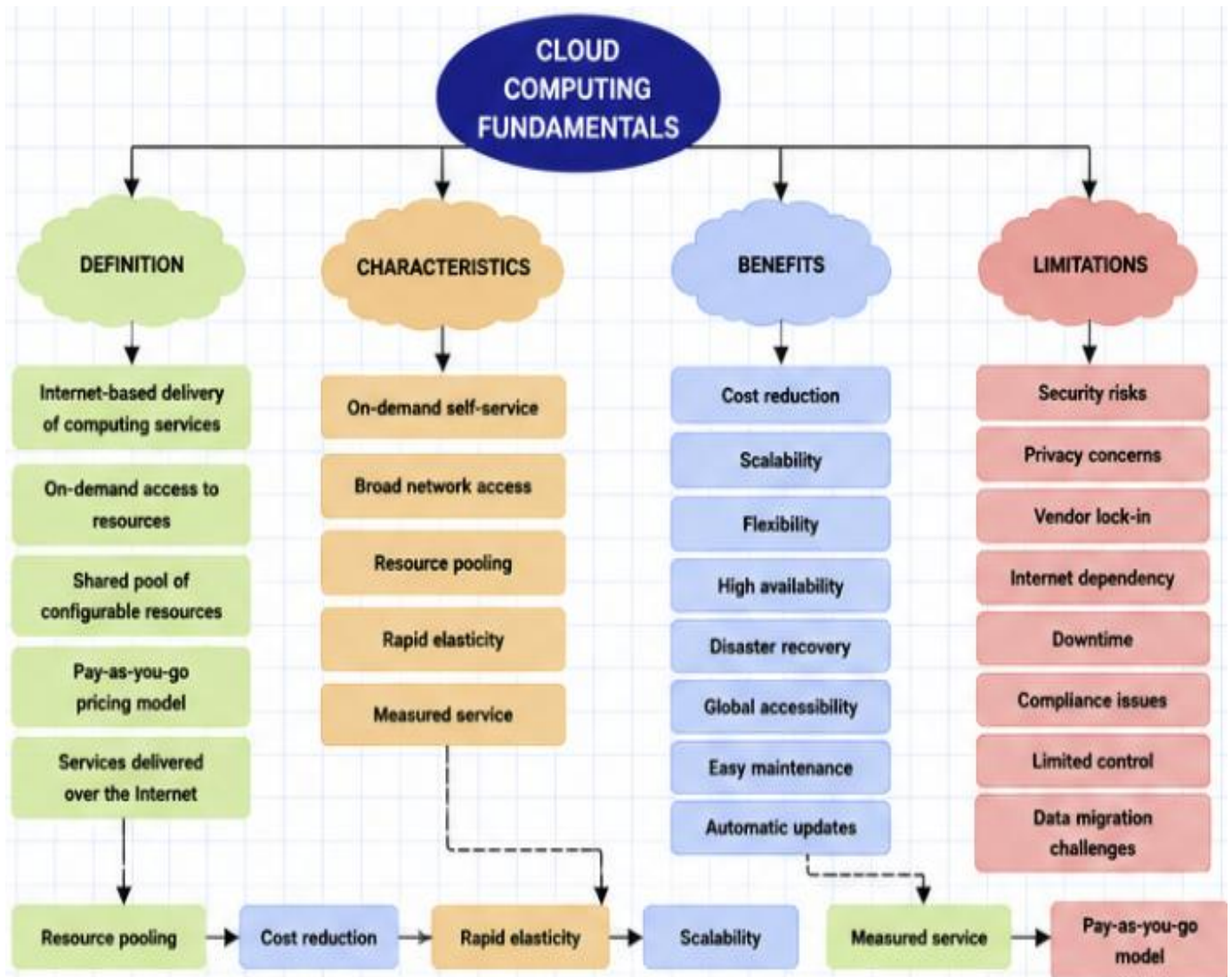
Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

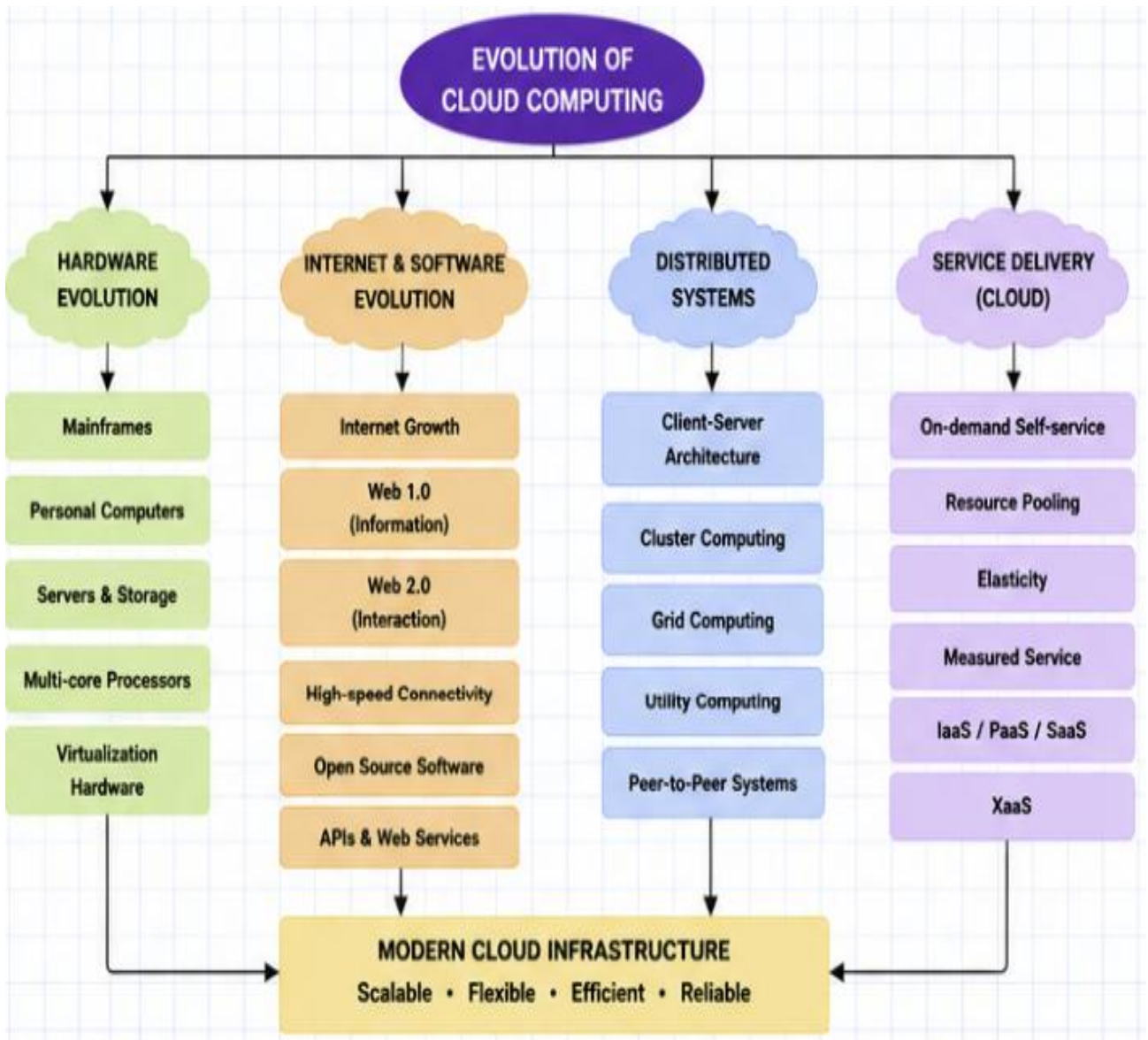
5. Questions

- i). Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?
- ii). Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?
- iii). Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?
- iv). Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?
- v). Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.



2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.



i). Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?

The concept map effectively represents the fundamental principles of cloud computing by organizing the topics into a clear hierarchical structure. The central concepts, such as Cloud Computing Fundamentals, Evolution of Cloud Computing, Service Models, and Virtualization & Web Services, are placed as the main nodes, while related subtopics are arranged beneath them in a logical sequence. This organization makes it easier to understand how each concept contributes to the overall cloud computing ecosystem.

Several design decisions improve the effectiveness of the concept map. Different colors are used to distinguish major categories, allowing related concepts to be identified quickly. Cloud shapes are used for major concepts and rounded rectangles for subtopics, creating a consistent visual hierarchy. Directional arrows clearly show relationships between concepts, while cross-links demonstrate dependencies, such as virtualization enabling cloud services and web services supporting cloud communication. These design choices improve readability, reduce confusion, and help learners understand both individual concepts and their interrelationships.

ii) Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?

Scalability, elasticity, and resource management are fundamental characteristics of cloud computing that directly influence cloud architecture. Scalability allows computing resources such as storage, processing power, and network capacity to increase or decrease according to user demand. Elasticity enables the automatic allocation and release of resources, ensuring that applications continue to perform efficiently during varying workloads. Resource management combines resource pooling, load balancing, and measured services to maximize hardware utilization while minimizing operational costs.

If these capabilities were absent, cloud architecture would become inefficient. Systems would struggle to handle increasing workloads, leading to slow response times, server overload, and service interruptions. Resources would remain underutilized during periods of low demand, increasing operational costs. Poor resource allocation could also reduce reliability, create downtime, and negatively affect the user experience. Therefore, scalability, elasticity, and resource management are essential for maintaining a reliable and efficient cloud environment.

iii) Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?

The concept map includes REST APIs, SOAP web services, and the cloud service models IaaS, PaaS, SaaS, and XaaS because they represent the essential technologies that enable communication, service delivery, and resource sharing in cloud computing.

REST was selected because it is lightweight, stateless, and widely used for modern web and mobile applications through HTTP and JSON. SOAP was included because it provides standardized XML-based communication with enhanced security features suitable for enterprise applications. APIs were chosen because they allow different applications and cloud services to communicate, exchange data, and integrate seamlessly.

The cloud service models provide different levels of cloud functionality. IaaS supplies virtual infrastructure such as servers, storage, and networking. PaaS provides development platforms and runtime environments for application developers. SaaS delivers complete software applications over the internet, while XaaS extends cloud services to

various specialized offerings. Together, these technologies improve flexibility, interoperability, automation, scalability, and efficient cloud service delivery.

iv) Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?

The relationships shown in the concept map illustrate how different cloud computing components work together to provide efficient and reliable cloud services. Virtualization creates virtual machines that support Infrastructure as a Service (IaaS), allowing multiple operating systems to share the same physical hardware efficiently. Web services, including REST and SOAP, communicate through APIs to enable secure and standardized data exchange between cloud applications.

These interactions improve reliability by minimizing downtime through dynamic resource allocation and redundancy. Performance is enhanced through load balancing, efficient resource utilization, and automatic scaling. User experience is improved because applications remain responsive, highly available, and accessible from multiple devices and locations. The interconnected relationships demonstrate that cloud computing functions as an integrated system rather than a collection of isolated technologies.

v) Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

Developing the concept map significantly improved my understanding of cloud computing architecture by helping me visualize how different concepts are interconnected. Rather than studying topics individually, I learned how cloud computing fundamentals, evolution, service models, virtualization, web services, and cloud characteristics work together to build a complete cloud ecosystem. The process also strengthened my ability to organize technical information logically and identify meaningful relationships between concepts.